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Relationships between Tropical Cyclone Survival Rates and Dynamical Equatorial Waves

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Key Findings

1. **More developers and higher survival rates** occurred in the **convergence and cyclonic vorticity phases** of **WMRG**, as well as the **cyclonic vorticity phases of ER1 and ER2**; the vice versa was true for opposite phases.
2. The percentage of developer in the favorable phases of **WRMG, ER1, and ER2** was significantly different from a random background at **12 days, 14 days, and 5 days** before TCG, respectively.

Introduction

1. Equatorial waves stronger influence tropical cyclone genesis (TCG) (Frank and Roundy 2006; Feng et al. 2023).
2. These waves have a period on seasonal-to-subseasonal (S2S) timescales, which may provide a source of S2S predictability for TCG (Judt 2020).
3. The link between dynamical equatorial waves and the evolution of TC seed remains unknown.

Research Question

- How do equatorial waves drive the evolution of tropical disturbance into TCG?

Implications

1. Equatorial waves serve as a potential source of S2S predictability for TCG.
2. Seeds in unfavourable phases at a single time step can still develop–evaluating the full trajectory is important.

Data and Method

- **Data**: JRA3Q and ERA5 reanalysis, 1980 – 2024, 24°N – 24°S, IBTrACS best track data
- **TC Tracking**: TRACK (Hodges et al. 2017) relative vorticity tracker:

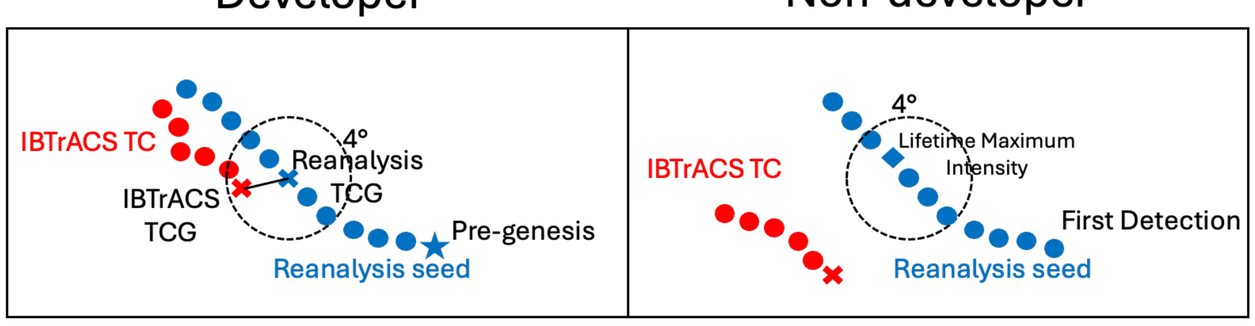


Fig. 1. Methodology for identify developers and non-developer in the reanalysis. Developers are reanalysis seeds that can be match spatiotemporally against IBTrACS TCs within 4° geodesic distance. Non-developers are those that cannot.

- **Dynamical equatorial wave detection:** Projected u, v, Z onto parabolic cylinder functions described by basic equatorial wave theory (Yang et al. 2003).
- **WMRG amplitude and phase**: Defined by convergence and vorticity and recorded at the longitude and time of each TC seed.

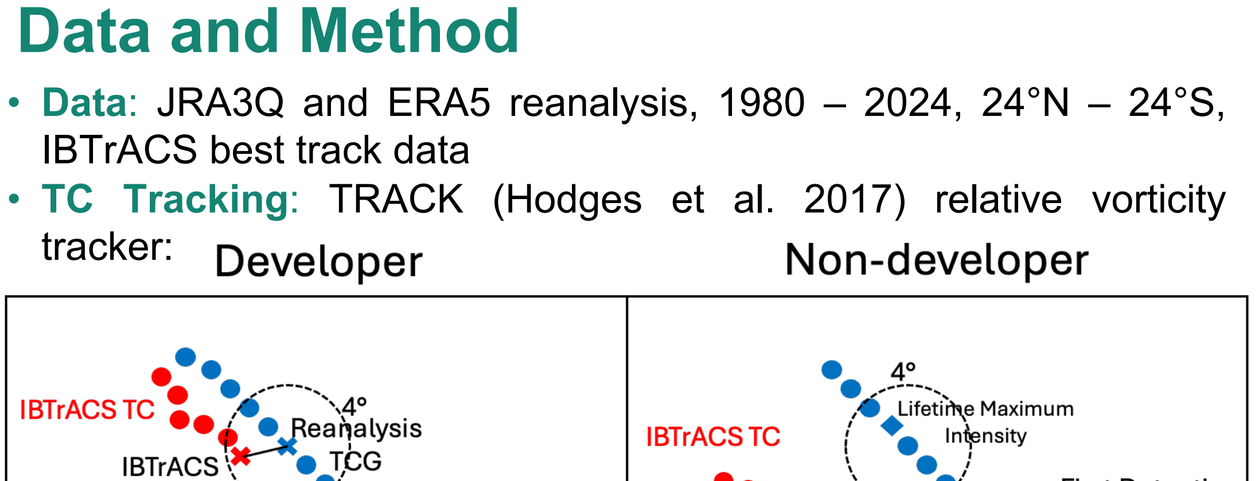


Fig. 2. Theoretical 850-hPa structure of WMRG fields (left) showing (shading; red indicates convergence) convergence, (contour; red means cyclonic) vorticity, and (vector) horizontal wind. The corresponding wave phase diagram (right). The track of TC Haiyan (2013) from (star) pre-genesis to (cross) TCG in terms of amplitude and phase is overlaid, with marker edge color representing WMRG amplitude (white = weakest).

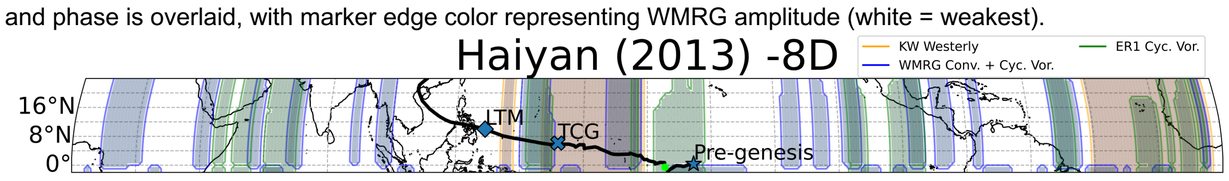


Fig. 3. Trajectory of TC Haiyan (2013) overlaid on (orange) KW westerly phases, (blue) WMRG convergence and cyclonic vorticity phases, and (green) ER1 cyclonic vorticity phases. The lime circle marks Haiyan's position 8 days before TCG.

Results

- Developers and TCG events significantly exceeded random background levels during favourable **WMRG** phases (**convergence and cyclonic vorticity**).
- This significance persisted up to **8 days before TCG.**
- Non-developers showed no preferred phase relationship.
- **SR** (developer / total seeds) is significantly higher during **favourable phases** of **strong** WMRG events.

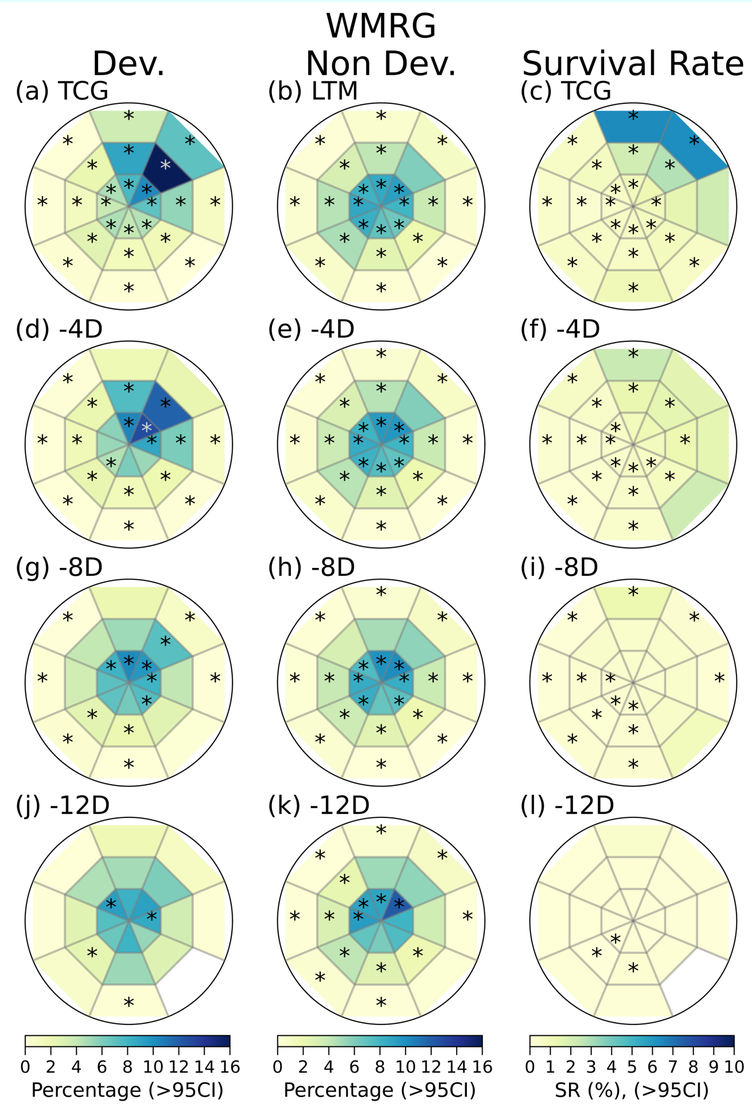
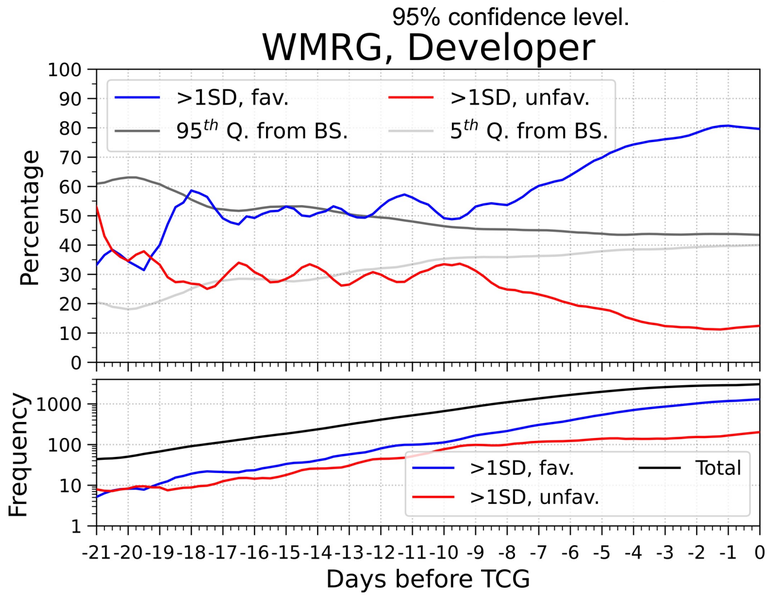


Fig. 4. WMRG wave phase diagrams showing percentages of (left) developers, (middle) non-developers, and (right) survival rate at (first row) TCG for developers and SR or lifetime maximum intensity for non-developers, (second to last row) and at 4, 8, and 12 days before TCG. Star indicate observed percentages that were significantly different from a bootstrapped random background distribution at the 95% confidence level.



- Favourable WMRG phases show significantly more developer up to **12.25 days prior to TCG** relative to random background expectation.

Fig. 5. Time series of developer percentages (top) for amplitude >1 SD WMRG during (blue) favorable and (red) unfavorable phases relative to the (grey) 95% bootstrap confidence interval. Developer frequencies in the favorable and unfavorable phase and (black) total developer counts are shown as a function of days before TCG (bottom).

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